

Must-Answer Computer Networks Interview Questions

Networking Fundamentals

1. What is a computer network?
2. Why are computer networks needed?
3. What are the advantages of computer networks?
4. What are the disadvantages of computer networks?
5. What is a node in a network?
6. What is a host in a network?
7. What is a client?
8. What is a server?
9. What is a peer-to-peer network?
10. What is a client-server network?
11. What is the difference between internet and intranet?
12. What is the difference between internet and extranet?
13. What is network topology?
14. What are the main types of network topologies?
15. What is bus topology?
16. What is star topology?
17. What is ring topology?
18. What is mesh topology?
19. What is tree topology?
20. What is hybrid topology?
21. What are the advantages of star topology?
22. What are the disadvantages of star topology?
23. What are the advantages of mesh topology?
24. What are the disadvantages of mesh topology?
25. What is LAN?
26. What is MAN?
27. What is WAN?
28. What is PAN?
29. What is CAN?
30. What is SAN?
31. What is WLAN?
32. What is VPN?
33. What is the difference between LAN and WAN?
34. What is bandwidth?
35. What is throughput?
36. What is the difference between bandwidth and throughput?
37. What is latency?
38. What is jitter?
39. What is packet loss?
40. What is propagation delay?
41. What is transmission delay?

42. What is queuing delay?
43. What is processing delay?
44. What is round-trip time?
45. What is bit rate?
46. What is baud rate?
47. What is the difference between bit rate and baud rate?
48. What is simplex communication?
49. What is half-duplex communication?
50. What is full-duplex communication?

OSI Model

51. What is the OSI model?
52. Why was the OSI model introduced?
53. How many layers are there in the OSI model?
54. What are the seven layers of the OSI model?
55. What is the role of the physical layer?
56. What is the role of the data link layer?
57. What is the role of the network layer?
58. What is the role of the transport layer?
59. What is the role of the session layer?
60. What is the role of the presentation layer?
61. What is the role of the application layer?
62. What is encapsulation in the OSI model?
63. What is decapsulation in the OSI model?
64. What is a PDU?
65. What is the PDU at the physical layer?
66. What is the PDU at the data link layer?
67. What is the PDU at the network layer?
68. What is the PDU at the transport layer?
69. What is the PDU at the application layer?
70. What is the difference between a frame, packet, segment, and bit?
71. Which OSI layer handles routing?
72. Which OSI layer handles error detection?
73. Which OSI layer handles end-to-end delivery?
74. Which OSI layer handles encryption and compression?
75. Which OSI layer handles application services?
76. Which OSI layer is responsible for MAC addressing?
77. Which OSI layer is responsible for IP addressing?
78. Which OSI layer is responsible for port numbers?
79. What devices operate at the physical layer?
80. What devices operate at the data link layer?
81. What devices operate at the network layer?
82. What is the difference between Layer 2 and Layer 3?
83. What is the difference between a Layer 2 switch and a Layer 3 switch?
84. Why is the OSI model useful for troubleshooting?
85. How does data flow from application layer to physical layer?

86. How does data flow from physical layer to application layer?
87. What is peer-to-peer communication between OSI layers?
88. What are service access points in OSI?
89. What is the difference between service and protocol?
90. What are common examples of protocols at each OSI layer?

TCP/IP Model

91. What is the TCP/IP model?
92. How many layers are there in the TCP/IP model?
93. What are the layers of the TCP/IP model?
94. What is the network access layer?
95. What is the internet layer?
96. What is the transport layer in TCP/IP?
97. What is the application layer in TCP/IP?
98. How does the TCP/IP model differ from the OSI model?
99. Why is TCP/IP more widely used than OSI?
100. Which TCP/IP layer corresponds to the OSI physical and data link layers?
101. Which TCP/IP layer corresponds to the OSI network layer?
102. Which TCP/IP layer corresponds to the OSI transport layer?
103. Which TCP/IP layer corresponds to the OSI session, presentation, and application layers?
104. What are the main protocols in the TCP/IP suite?
105. What is the role of IP in TCP/IP?
106. What is the role of TCP in TCP/IP?
107. What is the role of UDP in TCP/IP?
108. What is the role of ICMP in TCP/IP?
109. What is the role of ARP in TCP/IP?
110. What is the role of DNS in TCP/IP?
111. What is the role of DHCP in TCP/IP?
112. What is the role of HTTP in TCP/IP?
113. What is the role of FTP in TCP/IP?
114. What is the role of SMTP in TCP/IP?
115. What is the role of SSH in TCP/IP?
116. What is encapsulation in TCP/IP?
117. What is the difference between OSI encapsulation and TCP/IP encapsulation?
118. What is an internet protocol suite?
119. Why is TCP/IP called a protocol suite?
120. What happens when you send data over a TCP/IP network?

Physical Layer

121. What is the physical layer?
122. What are the responsibilities of the physical layer?
123. What is a transmission medium?
124. What are guided transmission media?
125. What are unguided transmission media?
126. What is twisted pair cable?

127. What is coaxial cable?
128. What is fiber optic cable?
129. What is the difference between single-mode and multi-mode fiber?
130. What are the advantages of fiber optic cable?
131. What are the disadvantages of fiber optic cable?
132. What is radio wave transmission?
133. What is microwave transmission?
134. What is infrared transmission?
135. What is attenuation?
136. What is distortion?
137. What is noise?
138. What is crosstalk?
139. What is signal-to-noise ratio?
140. What is modulation?
141. Why is modulation required?
142. What is amplitude modulation?
143. What is frequency modulation?
144. What is phase modulation?
145. What is digital modulation?
146. What is multiplexing?
147. What is frequency division multiplexing?
148. What is time division multiplexing?
149. What is wavelength division multiplexing?
150. What is code division multiplexing?
151. What is baseband transmission?
152. What is broadband transmission?
153. What is line coding?
154. What is Manchester encoding?
155. What is NRZ encoding?
156. What is scrambling?
157. What is a repeater?
158. What is a hub?
159. What is the difference between hub and switch?
160. Why are hubs rarely used today?

Data Link Layer

161. What is the data link layer?
162. What are the responsibilities of the data link layer?
163. What is framing?
164. Why is framing required?
165. What is a frame?
166. What is character stuffing?
167. What is bit stuffing?
168. What is flow control?
169. Why is flow control needed?
170. What is error control?

171. What is error detection?
172. What is error correction?
173. What is parity check?
174. What is checksum?
175. What is CRC?
176. Why is CRC widely used?
177. What is Hamming code?
178. What is stop-and-wait protocol?
179. What is sliding window protocol?
180. What is Go-Back-N ARQ?
181. What is Selective Repeat ARQ?
182. What is the difference between Go-Back-N and Selective Repeat?
183. What is piggybacking?
184. What is MAC?
185. What is MAC address?
186. What is the size of a MAC address?
187. What is the format of a MAC address?
188. What is unicast MAC address?
189. What is multicast MAC address?
190. What is broadcast MAC address?
191. What is MAC sublayer?
192. What is LLC sublayer?
193. What is CSMA?
194. What is CSMA/CD?
195. What is CSMA/CA?
196. Why is CSMA/CD used in Ethernet?
197. Why is CSMA/CA used in Wi-Fi?
198. What is collision?
199. What is collision domain?
200. What is broadcast domain?

Ethernet and Switching

201. What is Ethernet?
202. What is IEEE 802.3?
203. What is an Ethernet frame?
204. What fields are present in an Ethernet frame?
205. What is the minimum Ethernet frame size?
206. What is the maximum Ethernet frame size?
207. What is MTU?
208. What is jumbo frame?
209. What is a switch?
210. How does a switch learn MAC addresses?
211. What is a MAC address table?
212. What is CAM table?
213. What happens when a switch receives a frame with an unknown destination MAC?
214. What happens when a switch receives a broadcast frame?

- 215. What happens when a switch receives a frame for a known MAC address?
- 216. What is flooding in switching?
- 217. What is frame forwarding?
- 218. What is frame filtering?
- 219. What is store-and-forward switching?
- 220. What is cut-through switching?
- 221. What is fragment-free switching?
- 222. What is VLAN?
- 223. Why are VLANs used?
- 224. What is VLAN tagging?
- 225. What is IEEE 802.1Q?
- 226. What is access port?
- 227. What is trunk port?
- 228. What is native VLAN?
- 229. What is inter-VLAN routing?
- 230. What is a broadcast storm?
- 231. What is STP?
- 232. Why is Spanning Tree Protocol needed?
- 233. What is BPDU?
- 234. What is root bridge?
- 235. What is port cost in STP?
- 236. What are STP port states?
- 237. What is RSTP?
- 238. What is port security?
- 239. What is MAC flooding attack?
- 240. How can switching loops be prevented?

Network Layer

- 241. What is the network layer?
- 242. What are the responsibilities of the network layer?
- 243. What is logical addressing?
- 244. What is routing?
- 245. What is forwarding?
- 246. What is the difference between routing and forwarding?
- 247. What is an IP packet?
- 248. What are the fields of an IPv4 header?
- 249. What is TTL?
- 250. Why is TTL used?
- 251. What happens when TTL becomes zero?
- 252. What is fragmentation?
- 253. Why does IP fragmentation happen?
- 254. What is reassembly?
- 255. Where does IP packet reassembly occur?
- 256. What is DF bit?
- 257. What is MF bit?
- 258. What is path MTU discovery?

- 259. What is best-effort delivery?
- 260. Why is IP called unreliable?
- 261. What is a router?
- 262. What is a routing table?
- 263. What is next hop?
- 264. What is default gateway?
- 265. What is default route?
- 266. What is static routing?
- 267. What is dynamic routing?
- 268. What is source routing?
- 269. What is packet switching?
- 270. What is circuit switching?
- 271. What is message switching?
- 272. What is the difference between packet switching and circuit switching?
- 273. What is datagram packet switching?
- 274. What is virtual circuit packet switching?
- 275. What is congestion in a network?
- 276. What is congestion control at the network layer?
- 277. What is QoS?
- 278. What is traffic shaping?
- 279. What is traffic policing?
- 280. What is load balancing in networking?

IPv4 Addressing and Subnetting

- 281. What is an IP address?
- 282. What is IPv4?
- 283. What is the size of an IPv4 address?
- 284. What is dotted decimal notation?
- 285. What are the classes of IPv4 addresses?
- 286. What is Class A address?
- 287. What is Class B address?
- 288. What is Class C address?
- 289. What is Class D address?
- 290. What is Class E address?
- 291. What is classful addressing?
- 292. What is classless addressing?
- 293. What is subnetting?
- 294. Why is subnetting used?
- 295. What is subnet mask?
- 296. What is CIDR?
- 297. What is CIDR notation?
- 298. What is a network address?
- 299. What is a broadcast address?
- 300. What is a host address?
- 301. How do you calculate the number of hosts in a subnet?
- 302. How do you calculate the number of subnets?

- 303. What is a default subnet mask?
- 304. What is VLSM?
- 305. Why is VLSM used?
- 306. What is supernetting?
- 307. What is route aggregation?
- 308. What is private IP address?
- 309. What are the private IPv4 ranges?
- 310. What is public IP address?
- 311. What is loopback address?
- 312. What is 127.0.0.1 used for?
- 313. What is APIPA?
- 314. What is the APIPA address range?
- 315. What is 0.0.0.0 used for?
- 316. What is 255.255.255.255 used for?
- 317. What is limited broadcast?
- 318. What is directed broadcast?
- 319. What is a wildcard mask?
- 320. What is subnetting of 192.168.1.0/24 into four equal subnets?
- 321. How many usable hosts are available in a /24 network?
- 322. How many usable hosts are available in a /25 network?
- 323. How many usable hosts are available in a /26 network?
- 324. How many usable hosts are available in a /27 network?
- 325. How many usable hosts are available in a /28 network?
- 326. How many usable hosts are available in a /29 network?
- 327. How many usable hosts are available in a /30 network?
- 328. Why is /30 commonly used in point-to-point links?
- 329. What is a /31 subnet used for?
- 330. What is the difference between network ID and host ID?

IPv6

- 331. What is IPv6?
- 332. Why was IPv6 introduced?
- 333. What is the size of an IPv6 address?
- 334. How is an IPv6 address represented?
- 335. What is IPv6 address compression?
- 336. What is the loopback address in IPv6?
- 337. What is unspecified address in IPv6?
- 338. What is link-local address in IPv6?
- 339. What is global unicast address in IPv6?
- 340. What is multicast address in IPv6?
- 341. Does IPv6 support broadcast?
- 342. What replaces broadcast in IPv6?
- 343. What is anycast address?
- 344. What is SLAAC?
- 345. What is neighbor discovery protocol?
- 346. What is router advertisement?

- 347. What is router solicitation?
- 348. What is duplicate address detection?
- 349. What is IPv6 extension header?
- 350. What is the difference between IPv4 and IPv6 header?
- 351. What is dual stack?
- 352. What is tunneling in IPv6 transition?
- 353. What is NAT64?
- 354. What is 6to4 tunneling?
- 355. What are advantages of IPv6 over IPv4?
- 356. What are challenges in IPv6 adoption?
- 357. What is EUI-64?
- 358. What is IPv6 prefix length?
- 359. What is fe80::/10 used for?
- 360. What is ::1 used for?

ARP, ICMP, NAT, and DHCP

- 361. What is ARP?
- 362. Why is ARP needed?
- 363. How does ARP work?
- 364. What is ARP request?
- 365. What is ARP reply?
- 366. What is ARP cache?
- 367. What is gratuitous ARP?
- 368. What is proxy ARP?
- 369. What is ARP spoofing?
- 370. How can ARP spoofing be prevented?
- 371. What is RARP?
- 372. What is ICMP?
- 373. Why is ICMP used?
- 374. What is ping?
- 375. How does ping work?
- 376. What is traceroute?
- 377. How does traceroute work?
- 378. What ICMP message is used when TTL expires?
- 379. What is destination unreachable message?
- 380. What is echo request and echo reply?
- 381. What is NAT?
- 382. Why is NAT used?
- 383. What is static NAT?
- 384. What is dynamic NAT?
- 385. What is PAT?
- 386. What is NAT overload?
- 387. What is the difference between NAT and PAT?
- 388. What are the advantages of NAT?
- 389. What are the disadvantages of NAT?
- 390. What is port forwarding?

- 391. What is DHCP?
- 392. Why is DHCP used?
- 393. What is DORA process in DHCP?
- 394. What is DHCP Discover?
- 395. What is DHCP Offer?
- 396. What is DHCP Request?
- 397. What is DHCP Acknowledgement?
- 398. What is DHCP lease?
- 399. What is DHCP reservation?
- 400. What happens if DHCP server is unavailable?

Routing

- 401. What is routing?
- 402. What is a routing protocol?
- 403. What is an autonomous system?
- 404. What is administrative distance?
- 405. What is metric in routing?
- 406. What is hop count?
- 407. What is static route?
- 408. What is dynamic route?
- 409. What is default route?
- 410. What is directly connected route?
- 411. What is distance vector routing?
- 412. What is link state routing?
- 413. What is path vector routing?
- 414. What is the difference between distance vector and link state routing?
- 415. What is RIP?
- 416. What is the maximum hop count in RIP?
- 417. What is OSPF?
- 418. What is area in OSPF?
- 419. What is backbone area in OSPF?
- 420. What is DR in OSPF?
- 421. What is BDR in OSPF?
- 422. What is LSA in OSPF?
- 423. What is EIGRP?
- 424. What is BGP?
- 425. Why is BGP used on the internet?
- 426. What is iBGP?
- 427. What is eBGP?
- 428. What is route advertisement?
- 429. What is route convergence?
- 430. What is routing loop?
- 431. What is count-to-infinity problem?
- 432. What is split horizon?
- 433. What is route poisoning?
- 434. What is hold-down timer?

- 435. What is triggered update?
- 436. What is equal-cost multipath?
- 437. What is longest prefix match?
- 438. How does a router choose the best route?
- 439. What happens when multiple routes exist to the same destination?
- 440. What is route redistribution?
- 441. What is policy-based routing?
- 442. What is MPLS?
- 443. What is label switching?
- 444. What is GRE tunnel?
- 445. What is route summarization?
- 446. What is blackhole route?
- 447. What is floating static route?
- 448. What is recursive route lookup?
- 449. What is asymmetric routing?
- 450. How do routers prevent broadcast forwarding?

Transport Layer

- 451. What is the transport layer?
- 452. What are the responsibilities of the transport layer?
- 453. What is end-to-end communication?
- 454. What is a port number?
- 455. What is a socket?
- 456. What is a well-known port?
- 457. What is a registered port?
- 458. What is an ephemeral port?
- 459. What is TCP?
- 460. What is UDP?
- 461. What is the difference between TCP and UDP?
- 462. What is connection-oriented communication?
- 463. What is connectionless communication?
- 464. What is reliable delivery?
- 465. What is ordered delivery?
- 466. What is flow control in TCP?
- 467. What is congestion control in TCP?
- 468. What is TCP three-way handshake?
- 469. What is SYN?
- 470. What is SYN-ACK?
- 471. What is ACK?
- 472. Why is three-way handshake required?
- 473. What is TCP connection termination?
- 474. What is four-way handshake?
- 475. What is FIN?
- 476. What is RST?
- 477. What is TCP header?
- 478. What are TCP flags?

- 479. What is sequence number in TCP?
- 480. What is acknowledgment number in TCP?
- 481. What is TCP window size?
- 482. What is sliding window in TCP?
- 483. What is TCP slow start?
- 484. What is congestion avoidance?
- 485. What is fast retransmit?
- 486. What is fast recovery?
- 487. What is TCP timeout?
- 488. What is retransmission?
- 489. What is duplicate ACK?
- 490. What is Nagle's algorithm?
- 491. What is delayed ACK?
- 492. What is TCP keepalive?
- 493. What is UDP header?
- 494. Why is UDP faster than TCP?
- 495. When would you use UDP instead of TCP?
- 496. What applications use TCP?
- 497. What applications use UDP?
- 498. What is checksum in TCP and UDP?
- 499. What is multiplexing at the transport layer?
- 500. What is demultiplexing at the transport layer?

Application Layer Protocols

- 501. What is the application layer?
- 502. What are application layer protocols?
- 503. What is HTTP?
- 504. What is HTTPS?
- 505. What is the difference between HTTP and HTTPS?
- 506. What is HTTP request?
- 507. What is HTTP response?
- 508. What are common HTTP methods?
- 509. What is GET method?
- 510. What is POST method?
- 511. What is PUT method?
- 512. What is PATCH method?
- 513. What is DELETE method?
- 514. What is the difference between GET and POST?
- 515. What is idempotency in HTTP?
- 516. Which HTTP methods are idempotent?
- 517. What is an HTTP status code?
- 518. What are 1xx status codes?
- 519. What are 2xx status codes?
- 520. What are 3xx status codes?
- 521. What are 4xx status codes?
- 522. What are 5xx status codes?

- 523. What is 200 OK?
- 524. What is 201 Created?
- 525. What is 301 Moved Permanently?
- 526. What is 302 Found?
- 527. What is 400 Bad Request?
- 528. What is 401 Unauthorized?
- 529. What is 403 Forbidden?
- 530. What is 404 Not Found?
- 531. What is 500 Internal Server Error?
- 532. What is 502 Bad Gateway?
- 533. What is 503 Service Unavailable?
- 534. What is HTTP header?
- 535. What is cookie?
- 536. What is session?
- 537. What is cache-control header?
- 538. What is persistent connection in HTTP?
- 539. What is HTTP/1.1?
- 540. What is HTTP/2?
- 541. What is HTTP/3?
- 542. What is QUIC?
- 543. What is FTP?
- 544. What is the difference between FTP and SFTP?
- 545. What is SMTP?
- 546. What is POP3?
- 547. What is IMAP?
- 548. What is Telnet?
- 549. What is SSH?
- 550. Why is SSH preferred over Telnet?

DNS

- 551. What is DNS?
- 552. Why is DNS needed?
- 553. What is a domain name?
- 554. What is a fully qualified domain name?
- 555. What is DNS resolver?
- 556. What is recursive resolver?
- 557. What is root DNS server?
- 558. What is TLD server?
- 559. What is authoritative DNS server?
- 560. What is DNS zone?
- 561. What is DNS record?
- 562. What is A record?
- 563. What is AAAA record?
- 564. What is CNAME record?
- 565. What is MX record?
- 566. What is NS record?

- 567. What is TXT record?
- 568. What is PTR record?
- 569. What is SOA record?
- 570. What is SRV record?
- 571. What is DNS lookup?
- 572. What is recursive DNS query?
- 573. What is iterative DNS query?
- 574. What is forward DNS lookup?
- 575. What is reverse DNS lookup?
- 576. What is DNS caching?
- 577. What is TTL in DNS?
- 578. What is DNS propagation?
- 579. What is DNS spoofing?
- 580. What is DNS poisoning?
- 581. What is DNSSEC?
- 582. What port does DNS use?
- 583. Does DNS use TCP or UDP?
- 584. When does DNS use TCP?
- 585. What happens when you type a domain name in a browser?
- 586. How does a browser resolve a domain name?
- 587. What is split-horizon DNS?
- 588. What is round-robin DNS?
- 589. What is wildcard DNS record?
- 590. How do you troubleshoot DNS resolution problems?

Network Security

- 591. What is network security?
- 592. What are common network security threats?
- 593. What is malware?
- 594. What is phishing?
- 595. What is spoofing?
- 596. What is sniffing?
- 597. What is packet sniffing?
- 598. What is man-in-the-middle attack?
- 599. What is denial-of-service attack?
- 600. What is distributed denial-of-service attack?
- 601. What is SYN flood attack?
- 602. What is ARP spoofing attack?
- 603. What is DNS spoofing attack?
- 604. What is IP spoofing?
- 605. What is MAC spoofing?
- 606. What is firewall?
- 607. What is packet filtering firewall?
- 608. What is stateful firewall?
- 609. What is application firewall?
- 610. What is next-generation firewall?

- 611. What is IDS?
- 612. What is IPS?
- 613. What is the difference between IDS and IPS?
- 614. What is DMZ?
- 615. What is proxy server?
- 616. What is reverse proxy?
- 617. What is VPN?
- 618. What is site-to-site VPN?
- 619. What is remote access VPN?
- 620. What is IPsec?
- 621. What is SSL VPN?
- 622. What is TLS?
- 623. What is SSL?
- 624. What is the difference between SSL and TLS?
- 625. What is encryption?
- 626. What is symmetric encryption?
- 627. What is asymmetric encryption?
- 628. What is public key?
- 629. What is private key?
- 630. What is digital certificate?
- 631. What is certificate authority?
- 632. What is HTTPS certificate?
- 633. What is handshake in TLS?
- 634. What is zero trust security?
- 635. What is least privilege principle?
- 636. What is network segmentation?
- 637. What is access control list?
- 638. What is AAA in networking?
- 639. What is authentication?
- 640. What is authorization?
- 641. What is accounting?
- 642. What is RADIUS?
- 643. What is TACACS+?
- 644. What is port scanning?
- 645. What is vulnerability scanning?
- 646. What is penetration testing?
- 647. What is honeypot?
- 648. What is security policy?
- 649. How can a network be protected from unauthorized access?
- 650. What are common best practices for network security?

Wireless Networking

- 651. What is wireless networking?
- 652. What is Wi-Fi?
- 653. What is IEEE 802.11?
- 654. What is access point?

- 655. What is wireless router?
- 656. What is SSID?
- 657. What is BSSID?
- 658. What is wireless channel?
- 659. What is channel interference?
- 660. What is 2.4 GHz Wi-Fi?
- 661. What is 5 GHz Wi-Fi?
- 662. What is 6 GHz Wi-Fi?
- 663. What is the difference between 2.4 GHz and 5 GHz Wi-Fi?
- 664. What is Wi-Fi 4?
- 665. What is Wi-Fi 5?
- 666. What is Wi-Fi 6?
- 667. What is Wi-Fi 6E?
- 668. What is Wi-Fi 7?
- 669. What is WPA?
- 670. What is WPA2?
- 671. What is WPA3?
- 672. Why is WEP insecure?
- 673. What is wireless roaming?
- 674. What is hidden SSID?
- 675. Does hiding SSID improve security?
- 676. What is MAC filtering?
- 677. Is MAC filtering sufficient for wireless security?
- 678. What is captive portal?
- 679. What is ad hoc wireless network?
- 680. What is infrastructure wireless network?
- 681. What is MIMO?
- 682. What is beamforming?
- 683. What is Bluetooth?
- 684. What is Zigbee?
- 685. What is NFC?
- 686. What is cellular network?
- 687. What is 4G?
- 688. What is 5G?
- 689. What is handoff in wireless networks?
- 690. What causes weak Wi-Fi signal?

Cloud, Data Center, and Modern Networking

- 691. What is cloud networking?
- 692. What is virtual network?
- 693. What is VPC?
- 694. What is subnet in cloud networking?
- 695. What is internet gateway?
- 696. What is NAT gateway?
- 697. What is security group?
- 698. What is network ACL?

- 699. What is load balancer?
- 700. What is application load balancer?
- 701. What is network load balancer?
- 702. What is health check in load balancing?
- 703. What is horizontal scaling?
- 704. What is vertical scaling?
- 705. What is CDN?
- 706. Why is CDN used?
- 707. What is edge location?
- 708. What is reverse proxy caching?
- 709. What is API gateway?
- 710. What is service mesh?
- 711. What is sidecar proxy?
- 712. What is software-defined networking?
- 713. What is network function virtualization?
- 714. What is overlay network?
- 715. What is VXLAN?
- 716. What is container networking?
- 717. What is Kubernetes service networking?
- 718. What is ingress in Kubernetes?
- 719. What is egress traffic?
- 720. What is north-south traffic?
- 721. What is east-west traffic?
- 722. What is data center network?
- 723. What is spine-leaf architecture?
- 724. What is high availability in networking?
- 725. What is disaster recovery in networking?
- 726. What is multi-region networking?
- 727. What is hybrid cloud networking?
- 728. What is direct connect?
- 729. What is peering?
- 730. What is transit gateway?

Network Troubleshooting

- 731. How do you troubleshoot no internet connectivity?
- 732. How do you troubleshoot slow internet?
- 733. How do you troubleshoot high latency?
- 734. How do you troubleshoot packet loss?
- 735. How do you troubleshoot DNS failure?
- 736. How do you troubleshoot DHCP failure?
- 737. How do you troubleshoot duplicate IP address issue?
- 738. How do you troubleshoot default gateway issue?
- 739. How do you troubleshoot VLAN misconfiguration?
- 740. How do you troubleshoot routing issue?
- 741. How do you troubleshoot firewall blocking traffic?
- 742. How do you troubleshoot port connectivity?

- 743. How do you troubleshoot Wi-Fi connection problems?
- 744. How do you troubleshoot intermittent network disconnection?
- 745. How do you troubleshoot asymmetric routing?
- 746. How do you troubleshoot MTU mismatch?
- 747. How do you troubleshoot IP conflict?
- 748. How do you troubleshoot ARP issues?
- 749. How do you troubleshoot TCP connection failure?
- 750. How do you troubleshoot SSL certificate errors?
- 751. What is ping used for?
- 752. What is traceroute used for?
- 753. What is nslookup used for?
- 754. What is dig used for?
- 755. What is netstat used for?
- 756. What is ss command used for?
- 757. What is ipconfig used for?
- 758. What is ifconfig used for?
- 759. What is ip command used for?
- 760. What is route command used for?
- 761. What is arp command used for?
- 762. What is telnet used for in troubleshooting?
- 763. What is nc or netcat used for?
- 764. What is tcpdump used for?
- 765. What is Wireshark used for?
- 766. What is packet capture?
- 767. What should you check first when a website is not opening?
- 768. What should you check when only one user cannot access the network?
- 769. What should you check when all users cannot access the network?
- 770. What should you check when only one website is not opening?

Common Port Numbers

- 771. What port is used by HTTP?
- 772. What port is used by HTTPS?
- 773. What port is used by FTP control connection?
- 774. What port is used by FTP data connection?
- 775. What port is used by SSH?
- 776. What port is used by Telnet?
- 777. What port is used by SMTP?
- 778. What port is used by DNS?
- 779. What port is used by DHCP server?
- 780. What port is used by DHCP client?
- 781. What port is used by TFTP?
- 782. What port is used by POP3?
- 783. What port is used by IMAP?
- 784. What port is used by SNMP?
- 785. What port is used by LDAP?
- 786. What port is used by LDAPS?

- 787. What port is used by RDP?
- 788. What port is used by MySQL?
- 789. What port is used by PostgreSQL?
- 790. What port is used by MongoDB?
- 791. What port is used by Redis?
- 792. What port is used by NTP?
- 793. What port is used by SIP?
- 794. What port is used by SMB?
- 795. What port is used by Kerberos?
- 796. Why should common port numbers be remembered for interviews?
- 797. What is the difference between TCP port and UDP port?
- 798. Can TCP and UDP use the same port number?
- 799. What is port binding?
- 800. What is port forwarding?

Scenario-Based Interview Questions

- 801. What happens when you type google.com in a browser and press Enter?
- 802. What happens when a device connects to a Wi-Fi network?
- 803. What happens when a computer gets an IP address automatically?
- 804. What happens when two devices have the same IP address?
- 805. What happens when a router receives a packet for an unknown network?
- 806. What happens when a switch receives a frame for an unknown MAC address?
- 807. What happens when DNS server is unreachable?
- 808. What happens when default gateway is missing?
- 809. What happens when subnet mask is configured incorrectly?
- 810. What happens when ARP cache contains a wrong MAC address?
- 811. What happens when a TCP SYN packet is dropped?
- 812. What happens when a firewall blocks outbound DNS?
- 813. What happens when MTU is too large for a path?
- 814. What happens when TTL expires during packet forwarding?
- 815. What happens when DHCP lease expires?
- 816. How would you design a network for a small office?
- 817. How would you design a network for a college campus?
- 818. How would you design a secure network for a company?
- 819. How would you separate guest Wi-Fi from employee network?
- 820. How would you connect two branch offices securely?
- 821. How would you reduce latency for users in different countries?
- 822. How would you protect a public web server from attacks?
- 823. How would you identify whether a problem is DNS or network connectivity?
- 824. How would you check whether a remote server port is open?
- 825. How would you confirm whether packet loss is happening?
- 826. How would you choose between TCP and UDP for a new application?
- 827. How would you design a chat application network flow?
- 828. How would you design a video streaming network flow?
- 829. How would you design a file transfer system?
- 830. How would you debug a website that works on mobile data but not office Wi-Fi?

831. How would you debug a website that opens by IP address but not by domain name?
832. How would you debug a server that is reachable by ping but not by HTTP?
833. How would you debug a server that is reachable inside LAN but not from the internet?
834. How would you debug intermittent Wi-Fi disconnections?
835. How would you debug slow file downloads?
836. How would you debug high CPU usage on a network device?
837. How would you debug broadcast storm symptoms?
838. How would you debug routing loops?
839. How would you debug duplicate MAC address issues?
840. How would you debug a VPN connection failure?
841. How would you design IP addressing for 500 hosts?
842. How would you divide a /24 network into eight equal subnets?
843. How would you choose a subnet mask for 100 hosts?
844. How would you explain TCP three-way handshake to an interviewer?
845. How would you explain DNS resolution to an interviewer?
846. How would you explain subnetting to an interviewer?
847. How would you explain NAT to an interviewer?
848. How would you explain the difference between router and switch?
849. How would you explain why UDP is used for live streaming?
850. How would you explain why HTTPS is secure?